

Name: _____

Throughout, $X \in \mathbb{R}^{n \times p}$ is a data matrix: n observations in its rows, p features in its columns, and row i of X is $x_i^\top$.

1. List the steps of PCA.

[6 points]

List the steps of the PCA algorithm given X and a dimension $k < p$. **For full credit:** Describe each step both in words and in clear notation, and indicate the dimensions of every matrix involved.

In words	In notation
Center the columns: subtract the feature means.	$\bar{x} = \frac{1}{n} X^\top \mathbf{1}$, $\tilde{X} = X - \mathbf{1} \bar{x}^\top$, still $n \times p$
Form the scatter covariance matrix of the centered data.	$\tilde{X}^\top \tilde{X}$, $p \times p$
Eigendecompose covariance matrix.	$\tilde{X}^\top \tilde{X} = V D V^\top$ with V orthogonal $p \times p$, D diagonal, $\lambda_1 \geq \dots \geq \lambda_p \geq 0$
Keep the first k principal directions.	$V_k = [v_1 \dots v_k]$, the first k columns of V ; $p \times k$
Project the centered data onto principal directions to get scores.	$Z = \tilde{X} V_k$, so $z_i = V_k^\top \tilde{x}_i$ for each observation; Z is $n \times k$

2. What if the scales of the columns differ?

[4 points]

Recall the iris data set, which records four iris measurements, in centimeters, on each of $n = 150$ flowers. For reference, the four columns of the original iris data (all in cm) have respective variations of 0.69, 0.19, 3.12 and 0.58, and the first principal component direction is (0.36, -0.08, 0.86, 0.36).

Suppose the second column, sepal width, is recorded in *millimeters* instead:

	sepal length (cm)	sepal width (mm)	petal length (cm)	petal width (cm)
1	5.1	35.0	1.4	0.2
2	4.9	30.0	1.4	0.2
3	4.7	32.0	1.3	0.2
4	4.6	31.0	1.5	0.2
$\vdots$				

Contrast the first principal component that you would obtain by running PCA on the modified iris dataset with that of the original dataset. **For full credit:** Answer with both words and notation.

Solution. PCA seeks to maximize the variation

$$\sum_{i=1}^n (x_i^\top w)^2$$

over unit vectors w . Recording sepal width in millimeters multiplies the second entry of every observation by 10, e.g., $x_i^\top w$ becomes

$$x_{i1}w_1 + \mathbf{10} x_{i2}w_2 + x_{i3}w_3 + x_{i4}w_4.$$

So the contribution to the variation from the second feature grows by a factor of 100, and this will make the new PC1 close to $(0, 1, 0, 0)^\top$.

3. Bonus: What if the data matrix is not centered?

[2 points]

Suppose you run PCA on a data matrix X that has not been centered. If one of the column sums is large, what might the first principal direction end-up describing? **For full credit:** Answer with both words and notation.

Solution. Again, PCA seeks to maximize the sum of $(x_i^\top w)^2$ over directions w . Adding and subtracting $\bar{x}$ gives:

$$\sum_{i=1}^n (x_i^\top w)^2 = \underbrace{\sum_{i=1}^n ((x_i - \bar{x})^\top w)^2}_{\text{the variation along } w} + \underbrace{n(\bar{x}^\top w)^2}_{\text{squared mean score}}$$

So the issue is that, without centering, PCA might simply choose w to be the direction of $\bar{x}$.